

Statistical and Business Intelligence Analysis

Port Logistics | Renewable Energy Performance | 12 Analytical Questions

Document Reference	ADAFI15665 - Final Analytical Report
Organization	UV Netware
Submitted To	Raghav Shah, CFO - UV Netware
Subject	Adani Ports & SEZ - Statistical Analysis and Strategic Insights
Date	March 2026
Contents	12 Determine Questions with Answers, Data Analysis, and Final Conclusion
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Q1

Using the historical cargo throughput data (339 MMT in FY2023, 420 MMT in FY2024, and 450 MMT in FY2025), what is the forecasted cargo throughput for FY2026 and FY2027 using Excel forecasting methods?

Given Data:

Fiscal Year	Cargo Throughput (MMT)	Period (x)
FY2023	339	1
FY2024	420	2
FY2025	450	3

FORECAST.LINEAR Method (Excel):

The linear regression on the 3-year dataset gives a slope of **55.5 MMT per year** and an intercept of 292.0.

Formula applied: Forecast = Intercept + Slope × Period

Fiscal Year	Period (x)	Forecasted Cargo (MMT)	Lower Bound (-10%)	Upper Bound (+10%)
FY2026	4	514 MMT	463 MMT	565 MMT
FY2027	5	569.5 MMT	513 MMT	626 MMT

Management guidance for FY2026 is 505–515 MMT, which closely aligns with the FORECAST.LINEAR projection of 514 MMT, validating the model's accuracy.

Conclusion - Based on FORECAST.LINEAR applied to FY2023–FY2025 data, cargo throughput is projected at 514 MMT in FY2026 and 570 MMT in FY2027. The average annual growth rate is 55.5 MMT per year. Growth is decelerating (FY24: +24%, FY25: +7%), so a conservative planning figure of 505–515 MMT for FY2026 aligns with management guidance and is recommended as the operational base.

Q2

Using the monthly cargo dataset (Jan–Dec), what will be the expected cargo throughput for the next three months based on a time-series forecast?

Given Data — Monthly Cargo Throughput:

Month	MMT	Month	MMT	Month	MMT
January	38.6	May	41.8	September	39.4
February	36.4	June	39.7	October	41.0
March	40.1	July	37.5	November	40.5
April	35.9	August	42.2	December	41.9

Statistical Summary: Total = 475.0 MMT | Mean (μ) = 39.58 MMT | Min = 35.9 (April) | Max = 42.2 (August)

FORECAST.LINEAR Calculation: Slope = 0.3357 MMT/month | Formula: Forecast = 37.4015 + 0.3357 × Period

Forecast Month	Period (x)	Expected Cargo (MMT)	Method
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January 2026	13	41.77 MMT	FORECAST.LINEAR
February 2026	14	42.10 MMT	FORECAST.LINEAR
Forecast Month	Period (x)	Expected Cargo (MMT)	Method
March 2026	15	42.44 MMT	FORECAST.LINEAR

Conclusion - Based on the time-series FORECAST.LINEAR trend applied to 12 months of exact data, the expected cargo volumes for the next three months are: January 2026 = 41.77 MMT, February 2026 = 42.10 MMT, and March 2026 = 42.44 MMT. The positive slope of 0.336 MMT per month indicates a steady upward trend in cargo demand, with all three forecast months exceeding the current annual mean of 39.58 MMT.

Q3

Based on regional cargo volumes, does cargo throughput significantly differ between regions according to ANOVA analysis?

Hypothesis:

H0 (Null): Regional differences do NOT significantly affect cargo throughput

H1 (Alternative): Regional differences DO significantly affect cargo throughput

Decision Rule: Reject H0 if p-value < 0.05 (significance level $\alpha = 0.05$) **ANOVA**

Input - 3-Year Regional Data (Repeated Measures):

Region	FY2023 (MMT)	FY2024 (MMT)	FY2025 (MMT)	Group Mean
West	151	200	200	183.7
East	83	110	110	101.0
South	68	90	90	82.7
North	38	50	50	46.0

ANOVA Output — Excel Data Analysis ToolPak (Single Factor):

Source of Variation	SS	df	MS	F-Statistic	p-value	F Critical ($\alpha=0.05$)
Between Groups	30,519	3	10,173	32.48	0.0000789	4.07
Within Groups	2,505	8	313	-	-	-
Total	33,025	11	-	-	-	-

Conclusion - REJECT H0. The F-statistic of 32.48 substantially exceeds the critical value of 4.07, and the p-value of 0.0000789 is far below the 0.05 threshold. This provides statistically significant evidence at 99.99% confidence that regional differences significantly affect cargo throughput. The West region (200 MMT) handles four times the cargo of the North region (50 MMT) - a structural disparity confirmed by the analysis, not random variation.

Q4

Given the port-level cargo volumes (Mundra 180 MMT, Krishnapatnam 59 MMT, Dhamra 43 MMT, Gangavaram 37 MMT, Hazira 26 MMT), what percentage of total cargo throughput does each port contribute?

Given Data:

Total Cargo = 180 + 59 + 43 + 37 + 26 = **345 MMT**

Formula: Contribution % = (Port Cargo / Total Cargo) × 100

Port	Cargo (MMT)	Contribution (%)	Cumulative (%)
Mundra	180	52.17%	52.17%
Krishnapatnam	59	17.10%	69.28%
Dhamra	43	12.46%	81.74%
Gangavaram	37	10.72%	92.46%
Hazira	26	7.54%	100.00%

Conclusion - Mundra port contributes the highest share of total cargo throughput at 52.17%, making it the most significant port among the five listed. The top three ports - Mundra (52.17%), Krishnapatnam (17.10%), and Dhamra (12.46%) - together account for 81.74% of listed cargo. This concentration highlights Mundra's critical role in the network and the importance of infrastructure resilience at this facility.

Q5

Based on historical cargo throughput growth from 339 MMT to 450 MMT, what cargo volume should be expected over the next two fiscal years, and what additional port capacity should be planned to accommodate that growth?

Given Data:

Year	Cargo Throughput (MMT)
FY2023	339
FY2025	450

Calculation:

Total Growth = 450 – 339 = **111 MMT** over 2 years

Average Annual Growth = 111 ÷ 2 = **55.5 MMT per year**

Year	Expected Cargo (MMT)	Calculation	Additional Capacity Needed
FY2026	506 MMT	450 + 55.5 = 505.5 ≈ 506	+56 MMT
FY2027	562 MMT	506 + 55.5 = 561.5 ≈ 562	+56 MMT

Current installed capacity = 627 MMT. At this growth rate, capacity saturation is projected by FY2028. Expansion priorities include the Vizhinjam deep-water transshipment port, Haldia terminal (North region), and East coast berth expansion at Krishnapatnam and Dhamra.

Conclusion - Based on historical average growth of 55.5 MMT per year, cargo throughput is expected to reach approximately 506 MMT in FY2026 and 562 MMT in FY2027. To accommodate this growth, an additional handling capacity of approximately 55–60 MMT per year is required through expansion of terminals, berths, storage facilities, and cargo handling infrastructure across the port network.

Q6

Which months show the highest cargo throughput (35.9 – 42.2 MMT range), and how should port operations prepare for similar demand peaks in the future?

Monthly Cargo Analysis - Peak and Trough Identification:

Month	Cargo (MMT)	Classification
January	38.6	Normal
February	36.4	TROUGH
March	40.1	Normal
April	35.9	TROUGH
May	41.8	PEAK
June	39.7	Normal
July	37.5	Normal
August	42.2	PEAK
September	39.4	Normal
October	41.0	PEAK
November	40.5	Normal
December	41.9	PEAK

Operational Preparedness Strategy:

- Schedule all major maintenance, equipment overhauls, and infrastructure upgrades during trough months (February: 36.4 MMT and April: 35.9 MMT) to protect peak-month operational capacity.
- Pre-position maximum berth availability, handling equipment, and additional labour shifts during August, May, December, October, and November.
- Improve logistics coordination with shipping lines, rail operators, and road transport to ensure seamless cargo movement during peak periods.
- Implement real-time demand forecasting and cargo monitoring systems to provide operations teams with 2–3 weeks advance notice of approaching peak months.

Conclusion — The five peak months are August (42.2 MMT), December (41.9 MMT), May (41.8 MMT), October (41.0 MMT), and November (40.5 MMT). The two trough months are April (35.9 MMT) and February (36.4 MMT). Port operations should concentrate all planned maintenance in February and April, and maintain maximum operational readiness with pre-deployed resources during the five peak months identified above.

Q7

If the probability of cargo exceeding 43 MMT in a month is low but possible, what operational strategy should be implemented to manage future peak demand scenarios?

Normal Distribution Analysis:

Parameter	Value	Formula Used
Mean (μ)	39.5833 MMT	=AVERAGE(monthly data)
Standard Deviation (σ)	2.1170 MMT	=STDEV(monthly data)

Test Value (x)	43 MMT	As given in problem brief
Z-Score	1.6140	$= (43 - \mu) \div \sigma$
$P(X \leq 43)$ - Cumulative	94.67%	$= \text{NORMDIST}(43, \mu, \sigma, \text{TRUE})$
$P(X > 43)$ - Exceedance	5.33%	$= 1 - \text{NORMDIST}(43, \mu, \sigma, \text{TRUE})$

Probability Table — Various Cargo Thresholds:

Cargo Threshold	Z-Score	P(Exceed)	Expected Frequency
41 MMT	+0.67	25.1%	~3 months per year
42 MMT	+1.14	12.8%	~1.5 months per year
43 MMT	+1.61	5.33%	~0.6 months per year
44 MMT	+2.09	1.8%	~0.2 months per year

Operational Strategy for Peak Demand Management:

- Activate a contingency berth protocol when monthly throughput tracking crosses 40.5 MMT (approximately the 90th percentile) — this gives 10–14 days advance notice before a potential 43 MMT breach.
- Maintain reserve handling capacity and additional storage space at Mundra and Krishnapatnam at all times to absorb sudden cargo volume increases.
- Use flexible workforce scheduling with pre-agreed additional labour shifts that can be activated within 48 hours of surge signals.
- Formalise overflow agreements with nearby ports (Kandla, JNPT for West region) to reroute excess bulk cargo without commercial penalties during rare high-volume months.

Conclusion - With a Z-score of 1.61 and $P(X > 43 \text{ MMT}) = 5.33\%$, exceeding 43 MMT in any month is a low-probability but operationally meaningful event occurring approximately once every 19 months. A proactive surge protocol, pre-positioned resources, and overflow port agreements should be maintained as standard operational practice to manage such scenarios without disruption.

Q8

With plant efficiencies ranging from 70% to 72%, what operational improvements should be implemented to increase energy generation efficiency?

Current Renewable Energy Portfolio:

Energy Type	Capacity (MW)	Portfolio Share	Current Efficiency Range
Solar	11,005	70.8%	70–72%
Hybrid (Wind+Solar)	2,556	16.5%	70–72%
Wind	1,977	12.7%	70–72%
Total	15,538 MW	100%	~71% Average

Recommended Operational Improvements:

- **Predictive Maintenance:** Deploy AI-based monitoring of solar panel degradation, inverter performance, and wind turbine blade health. This can reduce unplanned downtime from 3–4% to below 1.5%, adding

1.5–2 percentage points to efficiency.

- **Smart Grid and Transmission Optimization:** Upgrade power transmission infrastructure and integrate battery storage at port facilities to reduce transmission and distribution losses, which currently account for 2–4% of generated energy.
- **Hybrid Portfolio Scheduling:** Optimize hybrid wind-solar plant operations to maximize solar-wind complementarity (solar peaks at midday, wind peaks at evening). Target capacity factors of 55–60% on the 2,556 MW hybrid portfolio.
- **Automated Solar Panel Cleaning:** Soiling losses from dust in port environments can reduce solar output by up to 5%. Automated cleaning systems at high-throughput ports like Mundra and Krishnapatnam should be deployed as priority.

Conclusion — To increase energy generation efficiency beyond 72%, the port network should implement predictive AI maintenance to reduce downtime, upgrade transmission infrastructure to minimize losses, optimize hybrid portfolio scheduling for maximum capacity factors, and deploy automated cleaning systems to prevent soiling losses. These combined improvements can move average plant efficiency from 71% to 73–74%, generating approximately 2.7 TWh of additional annual energy at no additional fuel cost.

Q9

Considering cargo growth, port performance, revenue distribution, and renewable capacity, what three strategic priorities should leadership focus on for the next five years?

Data Foundation:

Factor	Current Status	Strategic Implication
Cargo Growth	339 → 450 MMT (+33% in 2 years)	Capacity saturation by FY2028 at current rate
Port Performance	Mundra = 52% of major port cargo	Critical concentration risk requiring diversification
Revenue Distribution	North = Rs.2.40B/MMT (highest efficiency)	North expansion yields highest revenue ROI
Renewable Capacity	15.5 GW at ~71% efficiency	2.7 TWh additional generation potential at 73%

Priority 1 - Expand Port Capacity and Network Infrastructure

Cargo demand is growing at 55.5 MMT per year on average. Current installed capacity of 627 MMT will approach saturation by FY2028. Leadership must commission the Vizhinjam deep-water transshipment port, accelerate the Haldia terminal in the North region, and expand berths at Krishnapatnam and Dhamra on the East coast. Target: add 55–60 MMT of handling capacity annually to match demand growth and reduce Mundra's concentration below 40% of the network.

Priority 2 - Improve Operational Efficiency and Revenue Per MMT

The North region generates Rs.2.40B per MMT versus West's Rs.1.75B - a 37% premium. Growing North from 50 to 90 MMT by FY2030 generates Rs.96 billion in additional annual revenue. Deploy automation, AI cargo tracking, and digital berth scheduling to reduce vessel turnaround time and logistics costs. Replicate North's high-value cargo model (containerised goods and specialty bulk) in the South region, which currently achieves Rs.2.00B per MMT.

Priority 3 - Scale Renewable Energy to 20 GW and Enable Green Hydrogen

Expand from 15.5 GW to 20 GW by FY2030, achieving energy independence across the entire port network. Develop green hydrogen production at Mundra SEZ using surplus solar generation, creating a new revenue stream. This reduces per-tonne operational fuel costs by an estimated 8–12% and positions APSEZ as a sustainability leader in global port operations.

Conclusion - Three strategic priorities for the next five years: (1) Expand port capacity by 55–60 MMT per year to prevent FY2028 saturation and reduce Mundra concentration; (2) Grow North region from 50 to 90 MMT to capture the highest revenue efficiency in the network; (3) Scale renewable energy to 20 GW to achieve energy independence, reduce costs, and enable a green hydrogen business at Mundra SEZ.

Q10

Based on regional revenues (Rs.350B West, Rs.210B East, Rs.180B South, Rs.120B North) and cargo throughput, which region generates the highest revenue efficiency per MMT, and what strategic action should follow?

Revenue Efficiency = Revenue / Cargo Throughput

Region	Revenue (Rs. Billion)	Cargo (MMT)	Revenue per MMT (Rs.B)	Relative Efficiency
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West	350	200	1.75	Baseline
Region	Revenue (Rs. Billion)	Cargo (MMT)	Revenue per MMT (Rs.B)	Relative Efficiency
East	210	110	1.91	+9% vs West
South	180	90	2.00	+14% vs West
North	120	50	2.40	+37% vs West — HIGHEST

Strategic Actions:

- **Prioritize North Region Capex:** Every additional 10 MMT in North generates Rs.24B revenue versus Rs.17.5B for equivalent West region volume — a 37% revenue premium per tonne of cargo handled.
- **Focus on High-Value Cargo:** North's efficiency reflects containerized and specialty bulk cargo (fertilisers, agri-commodities). Replicate this cargo mix at South region ports to lift South's efficiency above Rs.2.10B per MMT.
- **Strengthen Long-Term Contracts:** Leverage North's established revenue premium to secure 5–10 year take-or-pay contracts with steel, cement, and fertiliser manufacturers in the Bengal-Odisha industrial corridor.
- **Improve Pricing Strategy:** Benchmark North region tariffs against South and East to ensure the premium is fully captured and not being eroded by below-market pricing.

Conclusion - The North region generates the highest revenue efficiency at Rs.2.40 Billion per MMT — 37% more than the West region. Strategic action: prioritize North region capacity expansion through the Haldia terminal, targeting growth from 50 to 90 MMT by FY2030. This single initiative generates more additional revenue per tonne of new cargo than any other regional investment in the network.

Q1
1**How should operational costs be optimized across regions to increase profitability per unit of cargo handled?****Formula: Profitability per MMT = (Revenue – Operational Cost) / Cargo Throughput**

Region	Rev/MMT (Rs.B)	Cost Optimization Strategy
West	1.75	AI berth scheduling, rail evacuation optimization, electric cargo equipment
East	1.91	Shift cargo mix toward containers and LNG; reduce dry bulk dependence
South	2.00	Scale volume at Kattupalli; route transshipment cargo via Vizhinjam
North	2.40	Grow throughput 50 to 90 MMT to spread fixed infrastructure costs

Universal Cost Levers Applicable Across All Regions:

- **Energy Transition:** Replace diesel-based cargo handling equipment with electric systems powered by APSEZ's 15.5 GW renewable capacity. Estimated reduction: 8–12% in cost per tonne of cargo handled.
- **Automation and Digitalization:** Deploy automated cargo tracking, AI-based vessel scheduling, and digital gate management to reduce labour costs per tonne and improve throughput velocity at all ports.
- **Multimodal Transport Optimization:** Optimize rail-sea-rail routes using the existing 127-train, 690 km rail network to reduce dwell time, fuel costs, and last-mile delivery expenses across all regions.
- **Benchmarking:** Use the North region's cost efficiency model as the internal benchmark and replicate its practices in West and East regions where cargo complexity and volume justify the investment.

Conclusion - Operational costs can be optimized across all regions through energy transition (electric handling equipment), automation (AI scheduling), and multimodal logistics optimization. The North region's high revenue efficiency at Rs.2.40B per MMT provides the internal benchmark. The universal lever with the highest impact is replacing diesel cargo handling equipment with electric systems powered by APSEZ's own renewable energy — this directly reduces per-tonne costs while supporting sustainability targets.

Q1
2**Which customer segment should be prioritized to increase cargo volumes?****Chi-Square Test - Customer Type vs Cargo Category:** χ^2 Statistic = 21.3841 | Degrees of Freedom = 1 | Critical Value = 3.841 | p-value $\approx$ 0.0000038 | Decision:**REJECT H0**

Customer type statistically significantly influences cargo category choice (99.99% confidence).

Observed Cargo Distribution:

Customer Type	Bulk Cargo	Container Cargo	Total	Bulk Preference
Industrial	240 (69%)	110 (31%)	350	Strong Bulk
Retail	70 (47%)	80 (53%)	150	Slight Container
Total	310 (62%)	190 (38%)	500	—

Port-Level Cargo Distribution:

Port	Cargo (MMT)	Contribution (%)	Primary Cargo Type
Mundra	180	52.17%	Mixed — Bulk and Containers

Krishnapatnam	59	17.10%	Dry Bulk
Port	Cargo (MMT)	Contribution (%)	Primary Cargo Type
Dhamra	43	12.46%	Dry Bulk / LNG
Gangavaram	37	10.72%	Iron Ore and Coal
Hazira	26	7.54%	Containers and LNG

Recommendation:

The Industrial and Energy sector should be prioritized to increase cargo volumes. This includes steel plants, power plants, oil and gas companies, cement manufacturers, and large industrial exporters. This segment accounts for 70% of current cargo volume, strongly prefers bulk cargo (69% bulk preference) which aligns with APSEZ's dominant infrastructure, and generates predictable long-term demand through multi-year supply agreements.

Simultaneously, container terminal capacity at Krishnapatnam and Kattupalli should be developed to capture the Retail segment's higher revenue per tonne, particularly as these ports serve South India's growing manufacturing and export economy.

Conclusion - Prioritize the Industrial and Energy customer segment (steel, power, oil and gas, cement, large exporters) as they generate the highest cargo volumes, maintain long-term contracts, and align with APSEZ's bulk terminal infrastructure strength — statistically confirmed by the Chi-Square test ($\chi^2 = 21.38$, $p < 0.001$). Concurrently, develop container capacity at East and South ports to capture the higher-margin Retail segment.

FINAL CONCLUSION - ADANI PORTS & RENEWABLE ENERGY STRATEGIC ANALYSIS

Adani Ports and Special Economic Zone (APSEZ) has demonstrated exceptional operational momentum — growing from 339 MMT in FY2023 to 450 MMT in FY2025, consolidating its position as India's dominant port operator with a 27% national cargo market share. This comprehensive statistical and business intelligence analysis, conducted across five analytical frameworks, delivers the following key findings and strategic insights.

What the Statistical Analysis Confirms

The statistical analysis highlights clear differences in cargo performance across the regions. The ANOVA results ($F = 32.48$, $p = 0.0000789$) show that the variation in cargo throughput between regions is statistically significant at 99.99% confidence, indicating that the gap between the West and North regions is not due to random variation but reflects a structural difference in the network. The Normal Distribution analysis further shows that monthly cargo throughput remains relatively stable with an average (mean = 39.58 MMT, $\sigma = 2.12$ MMT). However, there is still a small probability (5.33%) that monthly cargo may exceed 43 MMT, which suggests that ports should be operationally prepared for occasional demand surges. In addition, the Chi-Square test ($\chi^2 = 21.38$, $p \approx 0.0000038$) confirms that customer type significantly influences cargo demand. The results clearly indicate that industrial customers play a major role in driving bulk cargo movement, making the industrial sector an important focus area for future growth.

The Central Strategic Insight

One of the key insights from the analysis is the difference between cargo volume and revenue efficiency across regions. The West region handles the largest share of cargo (200 MMT, 44.4% of total), but it generates the lowest revenue efficiency (Rs.1.75B per MMT). On the other hand, the North region handles the smallest volume of cargo (50 MMT, 11.1%), yet it records the highest revenue efficiency (Rs.2.40B per MMT). This indicates that higher cargo volume does not always translate into higher revenue efficiency. Therefore, instead of focusing only on expanding cargo volume in the West region, greater strategic value may come from strengthening the North region. Expanding infrastructure through the Haldia terminal and developing ports in the North-East could help scale this high-efficiency region and improve overall network performance.

Capacity and Growth Outlook

Based on the historical trend, cargo throughput has been growing at an average rate of about 55.5 MMT per year. Following this pattern, the expected cargo volume is projected to reach 506 MMT in FY2026 and 562 MMT in FY2027, which is also in line with the management guidance of 505–515 MMT. The monthly time-series forecast further estimates cargo movement of 41.77 MMT in January 2026, 42.10 MMT in February, and 42.44 MMT in March, indicating a steady increase in demand. These projections suggest that the port network is likely to experience a sustained period of high cargo activity. Although the current installed capacity of 627 MMT provides some operational buffer, continued growth at this pace could push the system close to capacity limits by FY2028, making timely expansion of port infrastructure necessary.

Summary of Key Metrics and Targets

Dimension	Current Status (FY2025)	5-Year Target (FY2030)
Total Cargo Throughput	450 MMT	800+ MMT
Mundra Port Concentration	52.17% of major ports	Below 40% through network diversification
Average Revenue Efficiency	Rs.1.91B per MMT	Rs.2.20B per MMT through cargo mix shift
Renewable Energy Capacity	15,538 MW at ~71% efficiency	20,000 MW at 74%+ efficiency
North Region Contribution	50 MMT (11.1% of total)	90 MMT (target 15%+ of total)
Customer Mix	Industrial 70%, Retail 30%	Industrial 65%, Retail 35%

APSEZ is not merely growing in volume — it is building the logistics backbone of India's industrial future. The statistical evidence confirms the significance of its regional performance differentiation, customer segmentation patterns, and throughput stability. Executing the three strategic priorities — East coast expansion to reduce Mundra concentration, North region development to capture premium revenue efficiency, and scaling renewables to 20 GW for energy independence — with the same operational discipline that has driven 33% cargo growth in two years, positions APSEZ firmly on the path to achieving its vision of 1 billion MT of cargo by 2030.

Prepared by: UV Netware | Reference: ADAFI15665 | Date: March 2026 | Submitted to: Raghav Shah, CFO